Table 1: Correction payloads, paths, and tags used in the analysis.

Era	Correction	Payload / path	Key / tag
Run3_2022	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer22_22Sep2023_V4_MC; JEC data: Summer22_22Sep2023_V4_DATA; JER: Summer22_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22CDSep23-Summer22-NanoAODv12/latest/puWeights.json.gz	Collisions2022_355100_357900_eraBCD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22CDSep23-Summer22-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoAODv12/latest/jetvetomaps.json.gz	Summer22_23Sep2023_RunCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22CDSep23-Summer22-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2022EE	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer22EE_22Sep2023_V4_MC; JEC data: Summer22EE_22Sep2023_Run{}_V4_DATA, Summer22EE_22Sep2023_V4_DATA; JER: Summer22EE_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/puWeights.json.gz	Collisions2022_359022_362760_eraEFG_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/jetvetomaps.json.gz	Summer22EE_23Sep2023_RunEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22EFGSep23-Summer22EE-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2023	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer23Prompt23_V4_MC; JEC data: Summer23Prompt23_Run{}_V4_DATA, Summer23Prompt23_V4_DATA; JER: Summer23Prompt23_RunCv1234_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23CSep23-Summer23-NanoAODv12/latest/puWeights.json.gz	Collisions2023_366403_369802_eraBC_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23CSep23-Summer23-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoAODv12/latest/jetvetomaps.json.gz	Summer23Prompt23_RunC_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23CSep23-Summer23-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2023BPix	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jet_jerc.json.gz	JEC MC: Summer23BPixPrompt23_V4_MC; JEC data: Summer23BPixPrompt23_Run{}_V4_DATA, Summer23BPixPrompt23_V4_DATA; JER: Summer23BPixPrompt23_RunD_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/puWeights.json.gz	Collisions2023_369803_370790_eraD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoAODv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload

Continued on next page

Table 1 – continued

Era	Correction	Payload / path	Key / tag
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jetvetomaps.json.gz	Summer23BPixPrompt23_RunD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2024	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt24_V5_MC; JEC data: Summer24Prompt24_V5_DATA; JER: Summer24Prompt24_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/puWeights_BCDEFGHI.json.gz	Collisions24_BCDEFGHI_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing_V
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt24_RunBCDEFGHI_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2024_378981_386951_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2025	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt25_V3_MC; JEC data: Summer24Prompt25_V3_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/afs/cern.ch/work/v/vdamante/Hmm_newSkim/corrections/data/MUO/SF/Run3-25Prompt-Summer24-NanoAODv15/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt25_RunCDEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-25Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2025_391658_398903_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2026	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt26_V2_MC; JEC data: Summer24Prompt26_Run{}_V1_DATA, Summer24Prompt26_V1_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-26Prompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt26_RunBCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-26Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2026_401624_403937_golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium

Table 2: Selections and corrections applied during NanoAOD skimming, in execution order. "Filter" removes events; "object selection" only selects objects; "definition" only creates stored columns.

Step	Operation	Kind	Condition / definition	Scope	Correction state
1	Orthogonal luminosity split	Event filter	`OrthogonalEraTag == 0/1/2`, assigned deterministically with luminosity fractions 110/111/26 and seed 12345.	MC, Run3 2024/2025/2026	Before every physics correction; uses run, luminosity block, event and FullEventId.
2	Generator VBF phase space	Definition only	For selected DY inclusive/VBF-filtered samples, `GenVBFFilter` is computed from GenJet and GenPart. In the current implementation no `df.Filter` is executed.	MC DY, Run3 2024/2025/2026	Before corrections; currently no event rejection.
3	Base event weights	Weight definition	Luminosity, generator sign, pileup and cross-section weights; QCD-scale sums when configured.	MC only	Pileup correction is evaluated here, before object corrections and selections. Weights do not filter events.
4	Golden JSON	Event filter	Accept `run:luminosityBlock` pairs contained in the era-specific certified JSON.	Data only	First data correction/filter, before object corrections.
5	Muon ScaRe	Object correction	Muon momentum scale for data and MC; resolution smearing for MC. Both NanoAOD and beam-spot-constrained momenta are produced, with scale/resolution variations for MC.	Data and MC	Applied before every muon selection.
6	Muon FSR recovery	Object correction	Add an FSR photon when $10^{-4} < \Delta R(\mu, \gamma) < 0.5$, no matched electron, $p_T^{\gamma}/p_T^{\mu} \leq 0.4$, $dR/E_T^2 \leq 0.012$, and relative isolation criterion ≤ 1.8.	Data and MC	Applied after ScaRe. The nominal selection uses `Muon_pt_FSR_corr`.
7	JEC/JER	Object correction	Reapply JEC; apply JER to MC; construct nominal and JER/JES Total shifted jet four-vectors when variations are enabled.	Data and MC (JER: MC)	Applied before MET flags and before all jet definitions/selections.
8	MET quality flags	Event filter	Logical AND of goodVertices, globalSuperTightHalo2016, EcalDeadCellTriggerPrimitive, BadPFMuon, BadPFMuonDz, hfNoisyHits, eeBadSc and ecalBadCalib. For specified 2022/2023 data runs the ecal flag is redefined with the dedicated bad-MET condition.	Data and MC	After object corrections; flags themselves are NanoAOD event flags (special bad-MET expression uses raw PuppiMET/Jet quantities).
9	Single-muon trigger matching	Event filter	`HLT_IsoMu24` and at least one muon with corrected+FSR $p_T > 26$ GeV matched to a trigger object satisfying ID 13 and filter bit 8 within $\Delta R < 0.4$.	Data and MC	Uses ScaRe+FSR muons. With MC variations enabled, the filter is an OR across nominal and shifted matching flags.
10	Good muon definition	Object selection	$p_T^{\mathrm{ScaRe+FSR}} > 15$ GeV, $\|\eta\| < 2.4$, medium ID, and `pfIsoId`$\geq$ 2; indices are sorted by corrected p_T.	Data and MC	Uses ScaRe+FSR momentum; also built for muon scale/resolution shifts in MC.
11	Exactly two muons	Event filter	Exactly two good muons. With variations enabled, exactly two are required for nominal and for every muon scale/resolution shift.	Data and MC	After ScaRe, FSR and trigger matching.
12	Dimuon mass window	Event filter	$50 < m_{\mu\mu} < 200$ GeV. With variations enabled, the window must be passed by nominal and every muon scale/resolution shift.	Data and MC	Mass is built from the selected ScaRe+FSR muons.
13	Electron veto	Event filter	Reject events containing an electron with $p_T > 20$ GeV, $\|\eta\| < 2.5$, and `Electron_mvaIso_WP90`.	Data and MC	Uses NanoAOD electron kinematics; no electron energy correction is applied in this skim.
14	Muon ID/ISO/trigger SF	Weight definition	Correctionlib scale factors for the configured muon sources, evaluated for both selected muons; central and, when requested, up/down branches are stored.	MC only	Applied after all muon/electron event filters; changes weights, not event acceptance.
15	Jet ID and b-tag flags	Object definition	Tight jet ID; loose/medium b-tag flags within $\|\eta\| < 2.5$; event b-tag veto flag requires zero medium-tagged jets and fewer than two loose-tagged jets.	Data and MC	Uses JEC/JER-corrected jet four-vectors and era-specific b-tag working points. The b-tag event flag is stored, not applied as a skim filter.
16	Jet veto map	Object definition	Evaluate the era-specific jet veto map and remove map-vetoed jets in the later selected-jet mask. `apply_filter=False` is passed by `skim.py`.	Data and MC	Uses corrected jets. No direct event filter is applied.
17	Selected jets	Object selection	Corrected $p_T > 20$ GeV, $\|\eta\| < 4.7$, tight jet ID, outside the jet-veto map, and $\Delta R(j, \mu_1) > 0.4$, $\Delta R(j, \mu_2) > 0.4$. Era-dependent horn flags are also stored.	Data and MC	Uses JEC/JER jets and nominal ScaRe+FSR selected muons; repeated for JER/JES shifts.
18	VBF jet candidates	Definition only	Build the VBF pair from selected jets outside the horn and store `HasVBF` and the two indices.	Data and MC	Uses corrected/selected jets; no VBF event filter during skimming.
19	Snapshot	Output	Write surviving events and selected nominal/shifted columns to the skim ROOT file.	Data and MC	Final step. Mass regions and analysis categories are not evaluated here.

Table 3: Era-dependent skim selection settings.

Era	Muon / jet thresholds	Horn veto definition	Special behavior
Run3_2022	$p_T^{\mu} > 15$, $p_T^{j} > 20$ GeV; $\|\eta_{\mu}\| < 2.4$, $\|\eta_j\| < 4.7$	$\|\eta_j\| > 2.5$ and $p_T^{j} < 50$ GeV	Special ecal bad-calibration handling for data runs 362433–362760.
Run3_2022EE	Same	$\|\eta_j\| \geq 2.5$ and $p_T^{j} < 50$ GeV	No special bad-run interval configured.
Run3_2023	Same	$\|\eta_j\| \geq 2.5$ and $p_T^{j} < 50$ GeV	Special ecal bad-calibration handling for data runs 366727–367144.
Run3_2023BPix	Same	$\|\eta_j\| \geq 2.5$ and $p_T^{j} < 50$ GeV	No special bad-run interval configured.
Run3_2024	Same	$2.5 \leq \|\eta_j\| < 3.0$ and $p_T^{j} < 50$ GeV	Orthogonal MC luminosity split; NanoAOD v15.
Run3_2025	Same	Always false (no horn veto)	Orthogonal MC luminosity split; NanoAOD v15.
Run3_2026	Same	Always false (no horn veto)	Orthogonal MC luminosity split; NanoAOD v15.

Important scope note. The `muons_selection`, `categories`, and `masses_regions` expressions in `config/Run3_*/selections.yaml` are not invoked by the current `analysis/skim.py`. Final baseline, VBF/ggF categories, b-tag veto, and analysis mass regions are therefore defined downstream at histogram level, not used to reject NanoAOD events during skimming.